Gauss-Jordan Elimination

Exercise 1:

$$x - 2y + 3z = 9$$
$$-x + 3y = -4$$
$$2x - 5y + 5z = 17$$

Solution 1:

LU Decomposition with Gaussian Elimination

Exercise

$$2x + y + 2z = -10$$
$$x + 2y + z = 8$$
$$3x + y - z = 2$$

Crout Method

Exercise:

$$2x + y + 2z = -10$$
$$x + 2y + z = 8$$
$$3x + y - z = 2$$

Jacobi

Exercise:

Solve the following linear system using Jacobi method. Set $\boldsymbol{X}^0 = 0$. Take $\varepsilon = 0.05$.

$$x_1 - 3x_2 + 12x_3 = 31$$
$$4x_1 + x_2 - x_3 = 3$$
$$2x_1 + 7x_2 + x_3 = 19$$

Solution:

Gauss-Seidel

Exercise:

Solve the following linear system using Gauss-Seidel method. Set $\boldsymbol{X}^0 = 0$. Take $\varepsilon = 0.05$.

$$x_1 - 3x_2 + 12x_3 = 31$$
$$4x_1 + x_2 - x_3 = 3$$
$$2x_1 + 7x_2 + x_3 = 19$$

Solution: